



# **TSMC 0.18um USB 2.0 I/O Library**

***tpd973g\_usb2***

**Data Sheet**

**v250b**

**2005/12/09**

***Taiwan Semiconductor Manufacturing Co., Ltd.***

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# **TSMC 0.18um USB 2.0 I/O Library**

## ***tpd973g\_usb2***

### **Data Sheet**

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## REVISION HISOTRY

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| Version     | Date       | Special Note                                                                                                              |
|-------------|------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>020</b>  | 2002/09/20 | 1. Preliminary release, 8 bit interface                                                                                   |
| <b>050</b>  | 2002/10/28 | 1. Preliminary release, include 16 bit interface<br>2. 16-bit BIST for high-speed is provided                             |
| <b>060</b>  | 2002/12/18 | 1. Preliminary release, 8-bit/16-bit programmable interface<br>2. Both 8-bit/16-bit BIST for high-speed mode are provided |
| <b>100</b>  | 2003/02/22 | 1. Formal release, 8-bit/16-bit programmable interface                                                                    |
| <b>110</b>  | 2003/10/20 | 1. Modify truth table description for BISTON pin<br>2. Modify FSEL[1:0] setup for external reference clock selection      |
| <b>120</b>  | 2003/10/27 | 1. Fix typo error.                                                                                                        |
| <b>130</b>  | 2003/12/08 | 1. Fix typo error.<br>2. Add power pad description.<br>3. Delete absolute maximum rating table.                           |
| <b>250b</b> | 2005/12/9  | Provide dedicated datasheet for tpd973g_usb2_250b library                                                                 |



## 1. GENERAL INFORMATION

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This data sheet contains the technical information of TSMC 0.18um USB 2.0 transceiver macrocell (UTM), which complies with the “Universal Serial Bus (USB) Specification Revision 2.0” and “USB 2.0 Transceiver Macrocell Interface (UTMI) Specification”. USB is a popular serial-link standard for PC peripherals and consumer products. To meet different digital interface, library *tpd973g\_usb2* provides the programmable UTMI operating at 60MHz/30MHz for 8-bit/16-bit digital interface. According to USB 2.0 specification, there are low-speed (1.5M bits/s), full-speed (12M bits/s), and high-speed (480M bits/s) transceiver protocols. The UTM of *tpd973g\_usb2* is designed for **high-speed capable device** applications, so it provides both full-speed and high-speed mode operations.

The UTM encodes the serialized input data and packs it with SYNC and EOP (end of packet) according to the operation mode when transmitting data. When receiving data, the SYNC and EOP are stripped before the output data are decoded and de-serialized. The UTM of *tpd973g\_usb2* is suitable for stand-alone-chip PHY as well as SoC-integrated IP. To reduce the test cost of high-speed mode in mass production, TSMC UTM provides the flexible built-in self-test (BIST) circuit. This BIST covers all analog and digital circuits running at speed (480MHz), but the interface signals are slow (say 30MHz). So, users can load their test vectors on the fly and only have to monitor single pin for BIST fail.\* For better termination and signal quality, the full-speed-mode series resistors (Rs) are built internally. To meet the resistor accuracy spec, the 1.5K pull-up resistor is connected externally. Users who adopt TSMC UTM in their SoC design can save one PLL macro with the PLL480 output pin as clock source, which is an attractive feature for cost-sensitive, like consumer products.

This USB2.0 macrocell is highly modulized and easy to integrate with TSMC standard I/O libraries for the compatible power/ground bus and flexible bonding style – EZBOND™. The layout of the USB 2.0 macrocell is suitable for 4 /5 /6 metal layers physical design based on “TSMC 0.18um Logic Salicide 1.8V/3.3V Design Rule (T-018-LO-DR-001)”.

Users can refer this document for information of macro category, DC and AC specifications, block diagram, truth table, and the design kits availability. For the other application issues, please refer to the “*tpd973g\_usb2* application note v250b”. As to the test configuration and silicon measurements, please refer to the “*tpd973g\_usb2* silicon report v250b”.

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\* Please refer to “*tpd973g\_usb2* application note v250b” for more information.



## 2. MACRO CATEGORY

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Table 1 shows the macro category including macro name and descriptions of TSMC library *tpd973g\_usb2*. *PUSB20TM* contains digital and analog power/ground pads, so there is no need to add extra power/ground pads. The power cut cell is also included to isolate power/ground from digital ring while preserving whole chip ESD performance.

Table 1. Macro Category

| Macro Name             | Descriptions                                                                 |
|------------------------|------------------------------------------------------------------------------|
| <b><i>PUSB20TM</i></b> | USB 2.0 transceiver macrocell (UTM) with 8-bit/16-bit programmable interface |

## 3. FUNCTION DESCRIPTION

### 3.1 Block Diagram

Fig. 1 shows the block diagram of “dual speed” USB 2.0 transceiver macrocell (UTM) with 8-bit/16-bit uni-directional digital interface to SIE (Serial Interface Engine) controller. To obtain the bi-directional 16-bit interface, user can apply the bi-directional I/O as structured in page 38 of UTMI specifications. For more details, please refer to “*tpd973g\_usb2* application note v250b”.

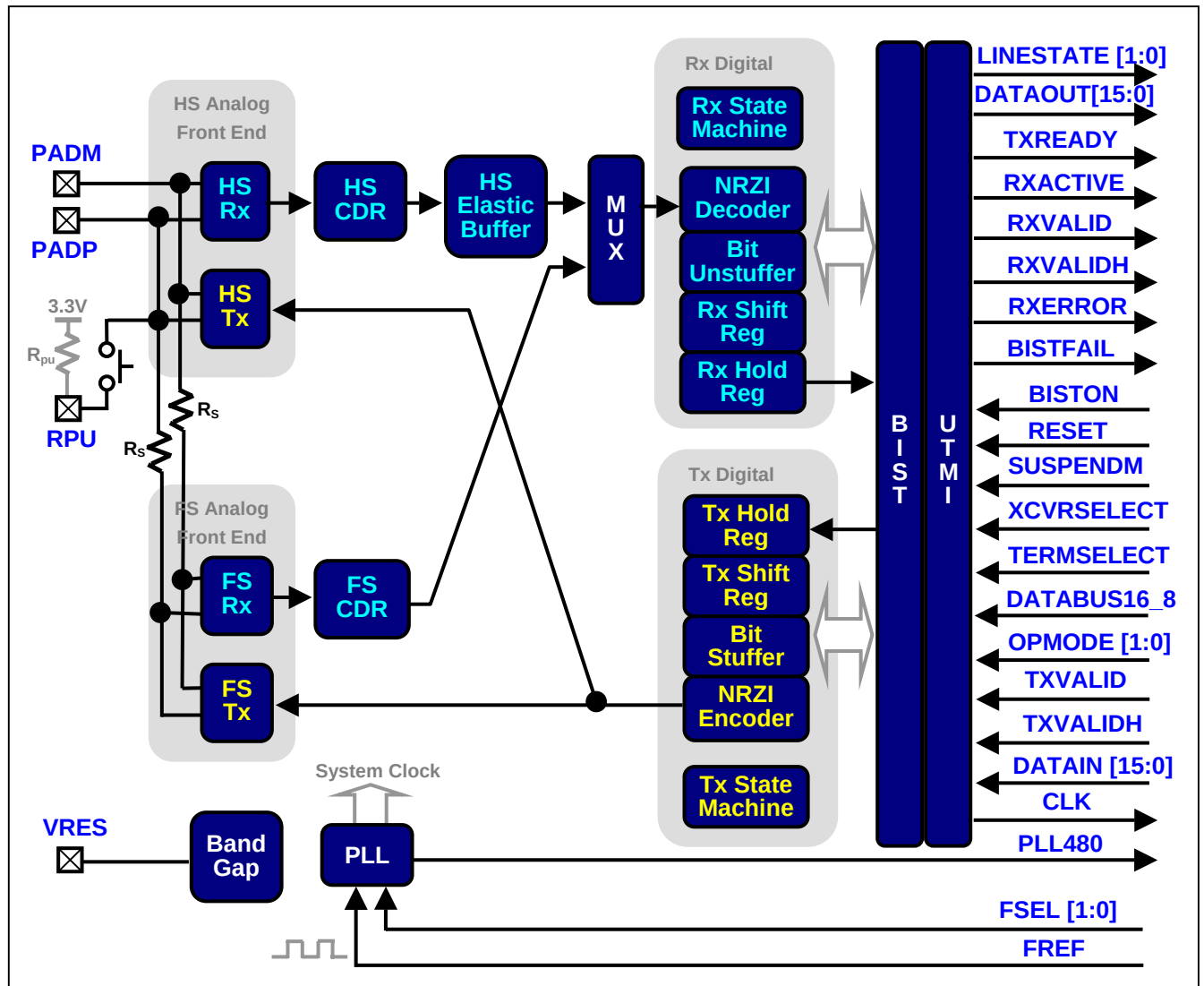


Fig. 2 Block diagram of USB 2.0 transceiver macrocell 16-bit interface



Table 2 describes the interface pins of the USB 2.0 transceiver macrocell.

Table 2. Pin Description of USB 2.0 Transceiver Macrocell

| Pin Name               | Pin Direction | Active Level | Pin Description                                                                                                               |
|------------------------|---------------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>CLK</b>             | Output        | ↑            | Output 30MHz clock for 16-bit HS/FS parallel interface to SIE or output 60MHz clock for 8-bit HS/FS parallel interface to SIE |
| <b>RESET</b>           | Input         | High         | Reset all state machine in UTM                                                                                                |
| <b>SUSPENDM</b>        | Input         | Low          | Place the macrocell in a mode drawing the minimum power<br>All current bias and clock are off when suspend active             |
| <b>XCVRSELECT</b>      | Input         | -            | Select between FS and HS transceiver:<br>0: HS transceiver enabled<br>1: FS transceiver enabled                               |
| <b>TERMSELECT</b>      | Input         | -            | Select between FS and HS terminations:<br>0: HS termination enabled<br>1: FS termination enabled                              |
| <b>DATABUS16_8</b>     | Input         | -            | Select between 16-bit interface and 8-bit interface<br>0: 8-bit interface enabled<br>1: 16-bit interface enabled              |
| <b>OPMODE [1:0]</b>    | Input         | -            | Select between various operation modes:                                                                                       |
|                        |               |              | [1] [0] Mode                                                                                                                  |
|                        |               |              | 0 0 0: Normal operation                                                                                                       |
|                        |               |              | 0 1 1: Non-Driving                                                                                                            |
|                        |               |              | 1 0 2: Disable bit stuffing and NRZI encoding                                                                                 |
|                        |               |              | 1 1 3: Reserved                                                                                                               |
| <b>LINESTATE [1:0]</b> | Output        | -            | Reflecting current state of the single ended receivers DP (LINESTATE[0]) and DM (LINESTATE[1])                                |
|                        |               |              | DM DP Mode                                                                                                                    |
|                        |               |              | 0 0 0: SE0                                                                                                                    |
|                        |               |              | 0 1 1: "J" state                                                                                                              |
|                        |               |              | 1 0 2: "K" state                                                                                                              |
|                        |               |              | 1 1 3: SE1                                                                                                                    |
| <b>DATAIN [7:0]</b>    | Input         | -            | 8-bit/16-bit parallel USB data input bus                                                                                      |
| <b>DATAIN [15:8]</b>   | Input         | -            | Used in 16-bit parallel USB data input bus only                                                                               |
| <b>TXVALID</b>         | Input         | High         | TXVALID indicates the DATAIN[7:0] bus is valid                                                                                |
| <b>TXVALIDH</b>        | Input         | High         | TXVALIDH indicates the DATAIN[15:8] bus is valid                                                                              |
| <b>TXREADY</b>         | Output        | High         | TXREADY acknowledges SIE that UTM has downloaded DATAIN to hold register, and is ready for the new data on the bus            |
| <b>DATAOUT [7:0]</b>   | Output        | -            | 8-bit/16-bit parallel USB data output bus                                                                                     |
| <b>DATAOUT [15:8]</b>  | Output        | -            | Used in 16-bit parallel USB data output bus only                                                                              |
| <b>RXACTIVE</b>        | Output        | High         | RXACTIVE indicates the receive state machine has detected SYNC and is negated after a bit stuff error or EOP is detected      |
| <b>RXVALID</b>         | Output        | High         | RXVALID indicates the DATAOUT[7:0] bus has valid data                                                                         |
| <b>RXERROR</b>         | Output        | High         | RXERROR indicates a receive error is detected                                                                                 |



| Pin Name          | Pin Direction  | Active Level | Pin Description                                                             |
|-------------------|----------------|--------------|-----------------------------------------------------------------------------|
| <b>BISTON</b>     | Input          | High         | Enable high-speed mode built-in self test (BIST)                            |
| <b>BISTFAIL</b>   | Output         | High         | Indicate the UTM failed to pass BIST                                        |
| <b>PADP</b>       | Bi-directional | -            | USB data pin Data+                                                          |
| <b>PADM</b>       | Bi-directional | -            | USB data pin Data-                                                          |
| <b>FREF</b>       | Input          | -            | Reference clock input pin, selected by FSEL [1:0]                           |
| <b>FSEL [1:0]</b> | Input          | -            | Select pins to identify the external reference clock frequency              |
|                   |                |              | [1] [0] External reference clock frequency                                  |
|                   |                |              | 0 0 3MHz                                                                    |
|                   |                |              | 1 0 6MHz                                                                    |
|                   |                |              | 0 1 12MHz                                                                   |
|                   |                |              | 1 1 24MHz                                                                   |
| <b>PLL480</b>     | Output         | -            | 480MHz clock output from PLL of UTM                                         |
| <b>RPU</b>        | Bi-directional | -            | Connected to an external 1.5K Ohm pull-up resistor                          |
| <b>VRES</b>       | Bi-directional | -            | Connected to an external 6.195K Ohm resistor for band-gap reference circuit |

The RXERROR detects the following error conditions: (a) full-speed stuff error, (b) full-speed byte boundary align error except “dribble”<sup>†</sup> condition, (c) elastic buffer over run/under run, (d) high-speed byte boundary align error, (e) high-speed non-EOP stuff error.

To fit the multiple of 480MHz PLL output, the reference clock input (FREF) can be only one of 3MHz, 6MHz, 12MHz, and 24MHz with proper setting of select signal FSEL [1:0]. They are all in crystal fundamental oscillating mode.

User can take the PLL480 output as their system clock source with proper divider, which can save a system PLL in their chip design.

<sup>†</sup> “Dribble” is the extra bit that happens before full-speed EOP. However, the extra bit could happen in any place of the packet, which should be identified by the CRC in SIE controller.



Table 3 describes the power pads of the USB 2.0 transceiver macrocell.

**Table 3. Power Pad Description of USB 2.0 Transceiver Macrocell**

| Pin Name    | Supply voltage | PWR/ GND | Pin Description        |
|-------------|----------------|----------|------------------------|
| <b>VDDA</b> | 3.3V           | PWR      | 3.3V analog power pad  |
| <b>VDDD</b> | 3.3V           | PWR      | 3.3V digital power pad |
| <b>VDDL</b> | 1.8V           | PWR      | 1.8V digital power pad |
| <b>VSSA</b> | 0V             | GND      | Analog ground pad      |
| <b>VSDL</b> | 0V             | GND      | Digital ground pad     |

PUSB20TM macrocell includes four power pads and three ground pads. Four power pads are VDDD, VDDL and two VDDA. Three ground pads are one VSDL and two VSSA. All power/ground pads support the necessary power for PUSB20TM macrocell. Users don't have to pay extra effort to handle power supply issue. For pad location and detail application, please refer to "tpd973g\_usb2 application note v250b".

## 3.2 Truth Table

Table 4 shows the line state of each transceiver mode.

Table 4. Line State of Each Transceiver Mode

| Transceiver Mode       |                | Full Speed        | High Speed | Chirp            |
|------------------------|----------------|-------------------|------------|------------------|
| <b>XCVRSELECT</b>      |                | 1                 | 0          | 0                |
| <b>TERMSELECT</b>      |                | 1                 | 0          | 1                |
| <b>LINESTATE [1:0]</b> | <b>SE0</b>     | !RXDPFS & !RXDMFS | RXSQHS     | RXSQHS           |
|                        | <b>J State</b> | RXDPFS & !RXDMFS  | !RXSQHS    | !RXSQHS & RXDHS  |
|                        | <b>K State</b> | !RXDPFS & RXDMFS  | Invalid    | !RXSQHS & !RXDHS |
|                        | <b>SE1</b>     | RXDPFS & RXDMFS   | Invalid    | Invalid          |

Table 5 shows the truth table of full-speed transceiver. Here “x” represents “don’t care” and “-” stands for “unknown”. Note that to work with full-speed mode, BISTON must keep low (disable).

Table 5. Truth Table of Full-Speed Transceiver

| Mode                   | Input     |            |            |           |          |          | Output   |          |            |            |           |
|------------------------|-----------|------------|------------|-----------|----------|----------|----------|----------|------------|------------|-----------|
|                        | STDB<br>Y | TXENF<br>S | TXSEF<br>S | TXDF<br>S | PAD<br>P | PAD<br>M | PAD<br>P | PAD<br>M | RXDPF<br>S | RXDM<br>FS | RXD<br>FS |
| <b>FS<br/>Transmit</b> | 0         | 0          | 0          | 0         |          |          | 0        | 1        | 0          | 1          | 1         |
|                        |           |            | 0          | 1         |          |          | 1        | 0        | 1          | 0          | 1         |
|                        |           |            | 1          | 0         |          |          | 0        | 0        | 0          | 0          | 1         |
|                        |           |            | 1          | 1         |          |          | 1        | 1        | 1          | 1          | 1         |
| <b>FS<br/>Receive</b>  | 0         | 1          | X          | x         | 0        | 0        |          |          | 0          | 0          | -         |
|                        |           |            |            |           | 0        | 1        |          |          | 0          | 1          | 0         |
|                        |           |            |            |           | 1        | 0        |          |          | 1          | 0          | 1         |
|                        |           |            |            |           | 1        | 1        |          |          | 1          | 1          | -         |
| <b>Suspend</b>         | 1         | x          | X          | x         | 0        | 0        |          |          | 0          | 0          | 1         |
|                        |           |            |            |           | 0        | 1        |          |          | 0          | 1          | 1         |
|                        |           |            |            |           | 1        | 0        |          |          | 1          | 0          | 1         |
|                        |           |            |            |           | 1        | 1        |          |          | 1          | 1          | 1         |



Table 6 shows the truth tables of the high-speed USB transceiver. Here “x” represents “don’t care” and “-“ stands for “unknown”.

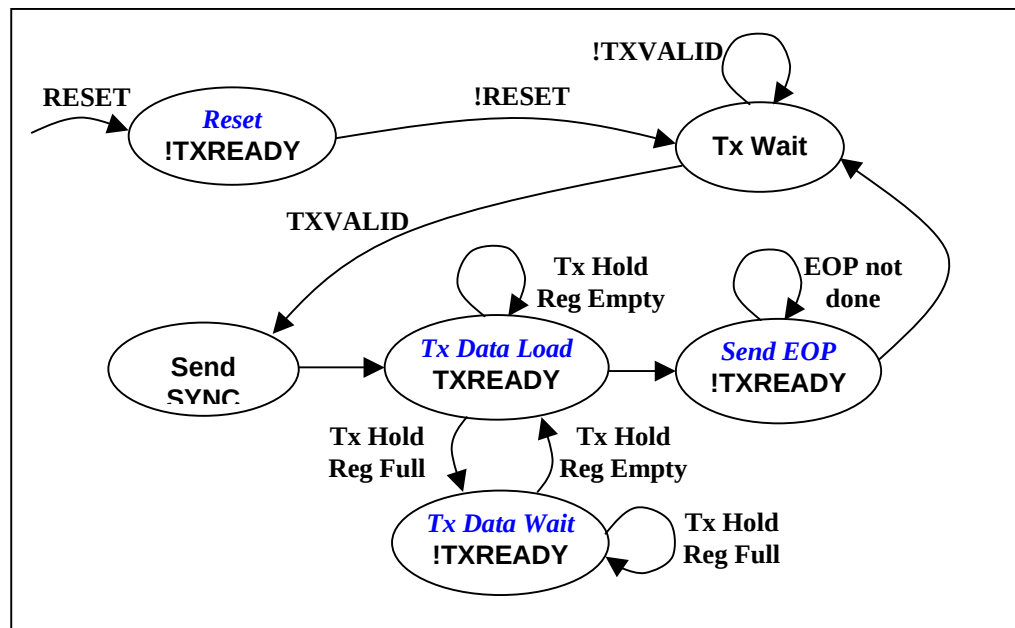
Table 6. Truth Table of High-Speed Transceiver

| Mode                   | Input |            |            |       |      |      | Output |      |            |       |
|------------------------|-------|------------|------------|-------|------|------|--------|------|------------|-------|
|                        | STDBY | BISTO<br>N | TXENH<br>S | TXDHS | PADP | PADM | PADP   | PADM | RXSQ<br>HS | RXDHS |
| <b>HS<br/>Transmit</b> | 0     | 0          | 0          | 0     |      |      | 0      | 1    | 1          | 1     |
|                        |       |            |            | 1     |      |      | 1      | 0    | 1          | 1     |
| <b>HS<br/>Receive</b>  | 0     | 0          | 1          | X     | 0    | 0    |        |      | 1          | 1     |
|                        |       |            |            |       | 0    | 1    |        |      | 0          | 0     |
|                        |       |            |            |       | 1    | 0    |        |      | 0          | 1     |
|                        |       |            |            |       | 1    | 1    |        |      | 1          | 1     |
| <b>HS<br/>BIST</b>     | 0     | 1          | 0          | 0     |      |      | 0      | 1    | 0          | 0     |
|                        |       |            |            | 1     |      |      | 1      | 0    | 0          | 1     |
|                        |       |            | 1          | X     | 0    | 0    |        |      | 1          | 1     |
|                        |       |            |            |       | 0    | 1    |        |      | 0          | 0     |
|                        |       |            |            |       | 1    | 0    |        |      | 0          | 1     |
|                        |       |            |            |       | 1    | 1    |        |      | 1          | 1     |
| <b>Suspend</b>         | 1     | x          | X          | x     | x    | x    |        |      | 1          | 1     |

### 3.3 Digital Function Description

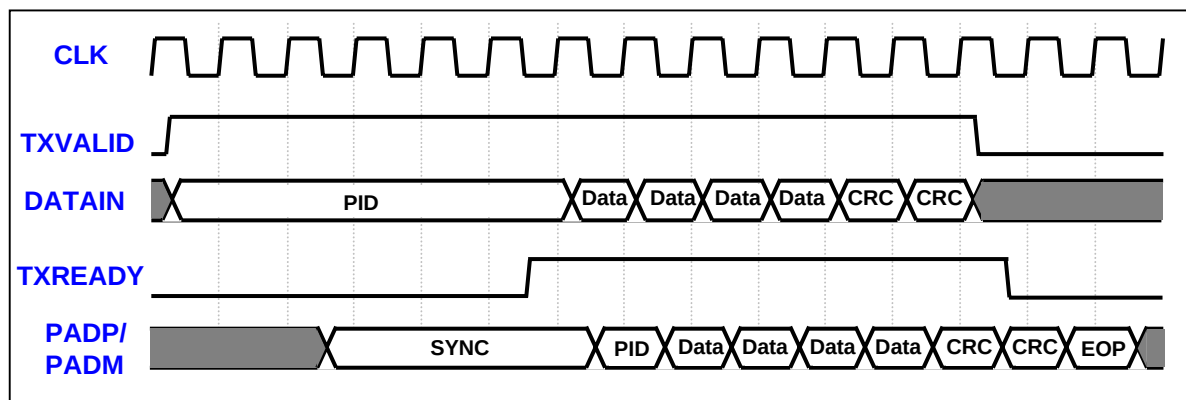
#### 3.3.1 Transmitter Digital Logics

Fig. 2 shows the transmitting state machine according to UTMI specification.



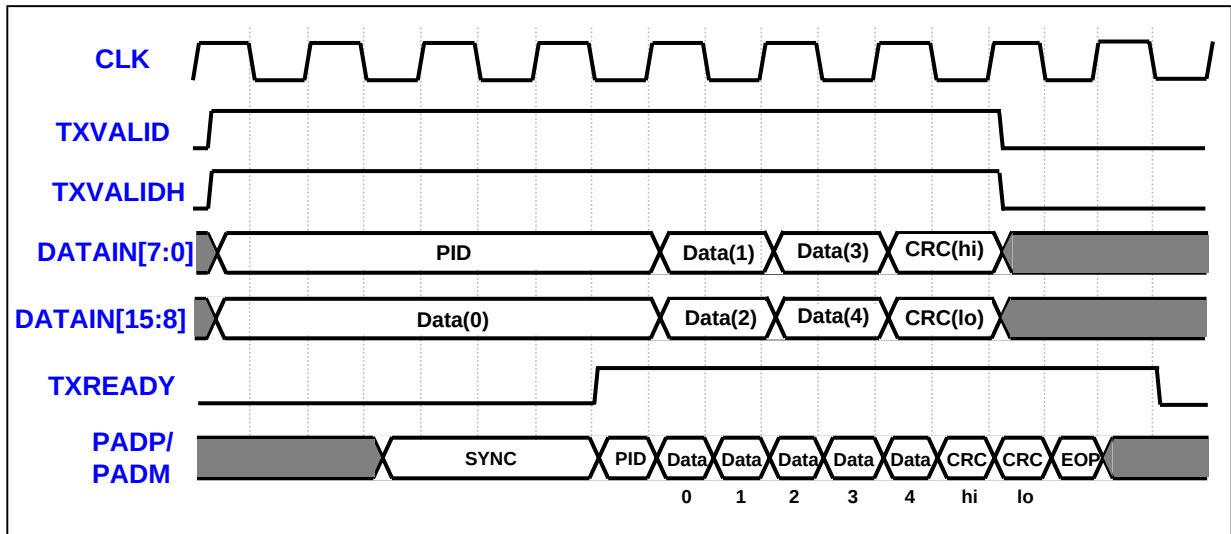
**Fig. 2 Transmitter State Diagram**

Fig. 3 shows the high-speed timing relationship between **TXVALID**, **DATAIN**, **TXREADY**, and the transmitting data (**PADP/PADM**) for 8-bit parallel interface. Here **CLK** is equal to 60 MHz.



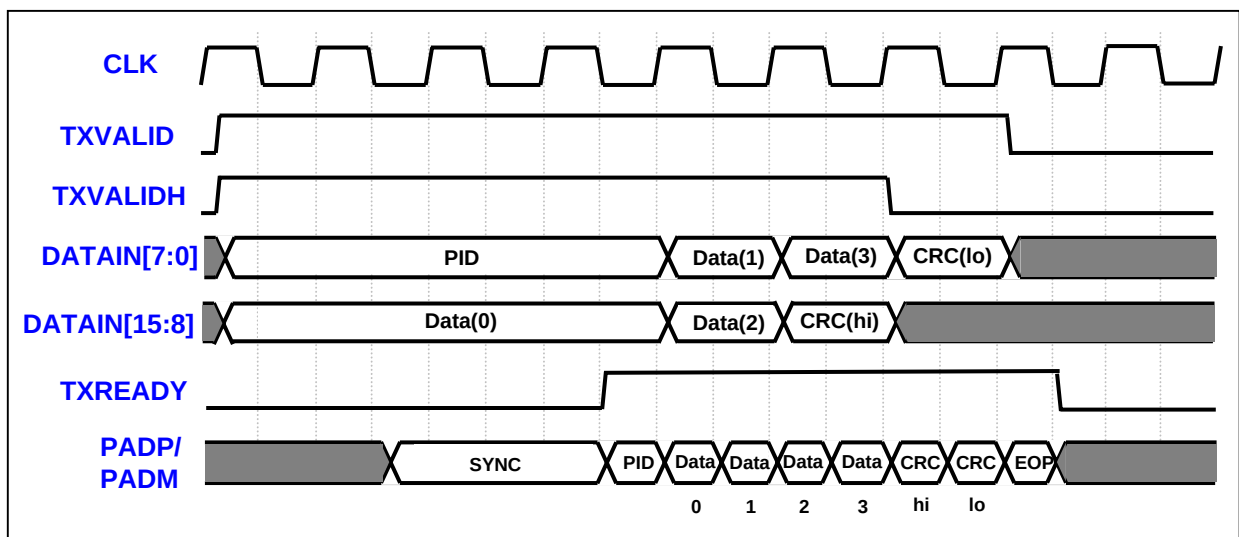
**Fig. 3 Transmitter Timing Diagram of 8-bit Data Packet**

Fig. 4 shows the high-speed timing relationship between TXVALID, TXVALIDH, DATAIN, TXREADY, and the transmitting data (PADP/PADM) for 16-bit parallel interface in even byte count case. Here CLK is equal to 30 MHz. Note that for OPMODE2 that requiring no SYNC and EOP patterns, only even byte transfer mode is supported.



**Fig. 4 Transmitter Timing Diagram of 16-bit Data Packet with Even Byte Count**

Fig. 5 shows the high-speed timing relationship between TXVALID, TXVALIDH, DATAIN, TXREADY, and the transmitting data (PADP/PADM) for 16-bit parallel interface in odd byte count case. Here CLK is equal to 30 MHz. Note that odd byte transfer is supported in OPMODE0 only.



**Fig. 5 Transmitter Timing Diagram of 16-bit Data Packet with Odd Byte Count**

### 3.3.2 Receiver Digital Logics

Fig. 6 shows the receiving state machine according to UTMI specification. Note that the error abortion is terminated only when the idle on the USB line is detected.

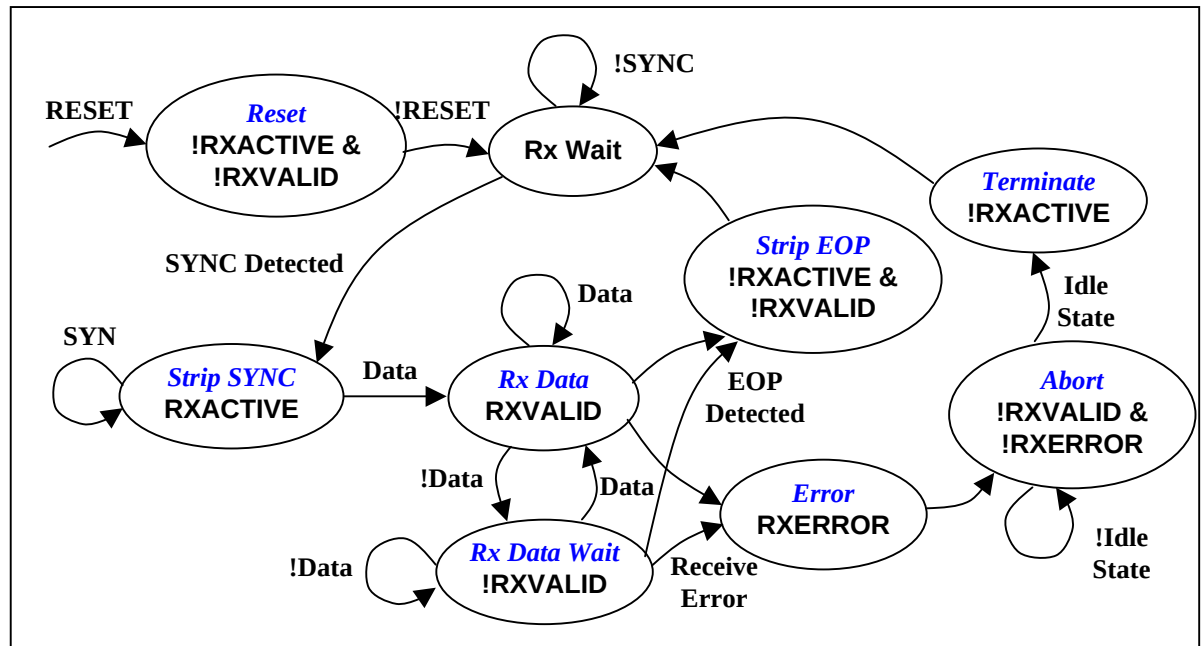


Fig. 6 Receiver State Diagram

Fig. 7 shows the high-speed timing relationship between RXACTIVE, DATAOUT, RXVALID, and the receiving data (PADP/PADM) for 8-bit parallel interface. Here CLK is equal to 60 MHz.

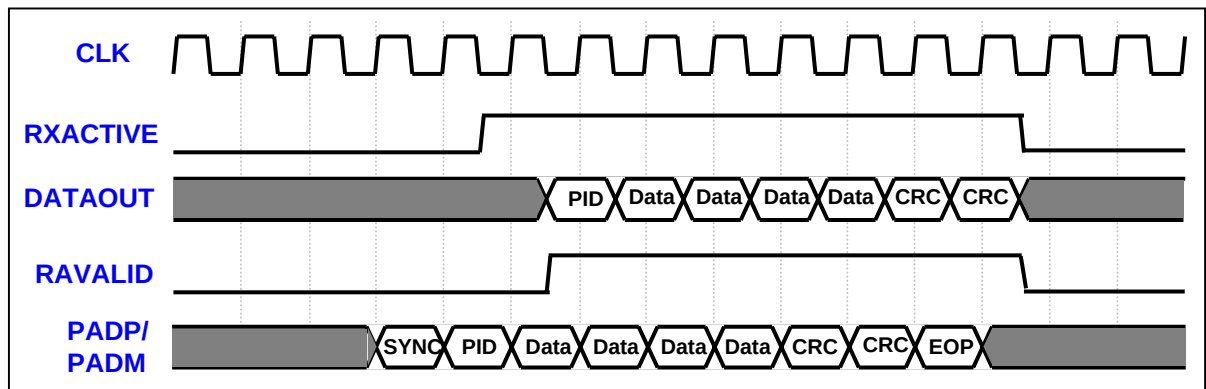
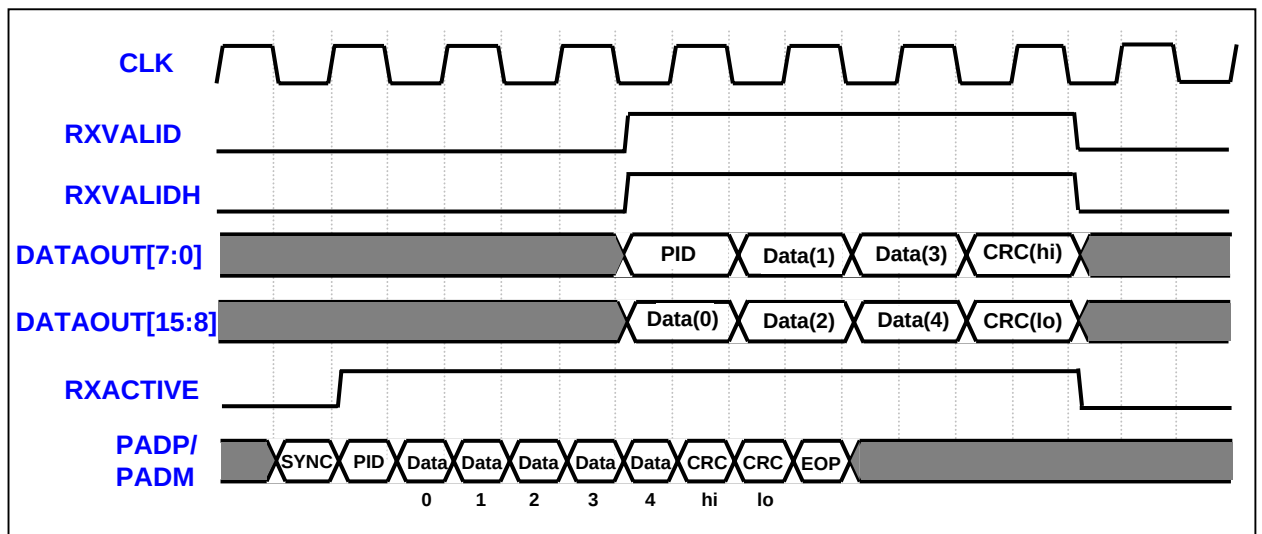


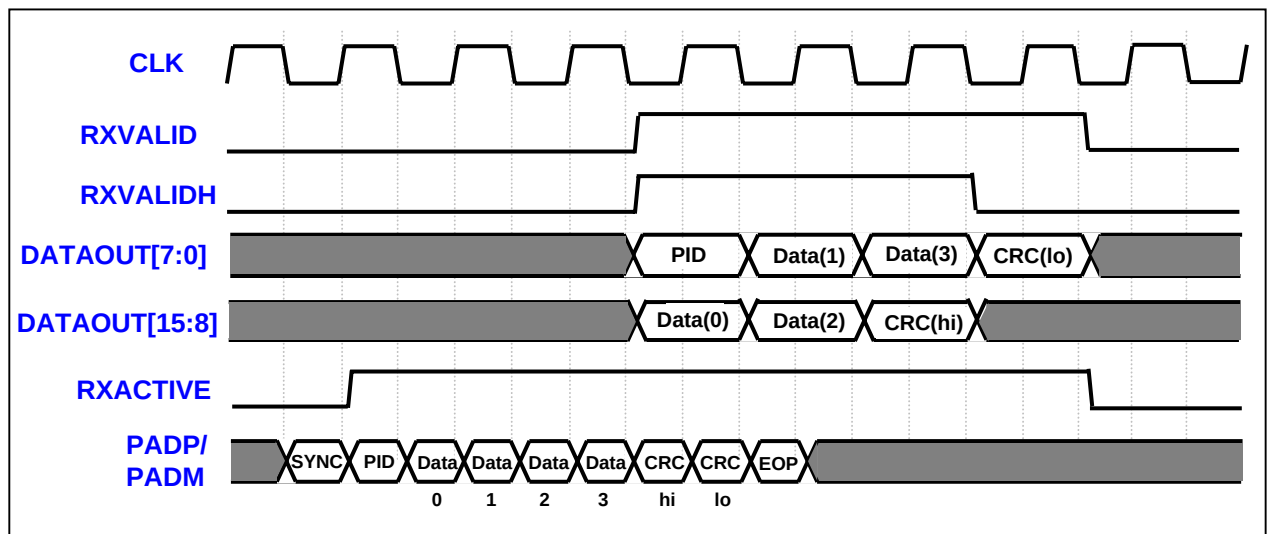
Fig. 7 Receiver Timing Diagram of 8-bit Data Packet

Fig. 8 shows the high-speed timing relationship between RXACTIVE, DATAOUT, RXVALID, RXVALIDH, and the receiving data (PADP/PADM) for 16-bit parallel interface. Here CLK is equal to 30 MHz.



**Fig. 8 Receiver Timing Diagram of 16-bit Data Packet with Even Byte Count**

Fig. 9 shows the high-speed timing relationship between RXACTIVE, DATAOUT, RXVALID, RXVALIDH, and the receiving data (PADP/PADM) for 16-bit parallel interface. Here CLK is equal to 30 MHz.



**Fig. 9 Receiver Timing Diagram of 16-bit Data Packet with Odd Byte Count**



## 4. ELECTRICAL SPECIFICATIONS

### 4.1 DC Characteristics

#### 4.1.1 DC Electrical Specifications

Table 7 and Table 8 summarize the full-speed and high-speed DC performance requirement of USB transceivers according to the Chapter 7 of “Universal Serial Bus Specification Version 2.0”. The USB transceiver of *tpd973g\_usb2* fully complies with the following specifications.

Table 7. Full-Speed DC Electrical Specifications

| Symbol              | Parameter                                      | Conditions             | Min             | Nom         | Max             |
|---------------------|------------------------------------------------|------------------------|-----------------|-------------|-----------------|
| $V_{IH}$            | Pad input high voltage                         |                        | 2.0V            |             |                 |
| $V_{IL}$            | Pad input low voltage                          |                        |                 |             | 0.8V            |
| $V_{DI}$            | Differential input sensitivity                 | FSDP-FSDM              | 0.2V            |             |                 |
| $V_{CM}$            | Common mode voltage range                      | Include $V_{DI}$ range | 0.8V            |             | 2.5V            |
| $V_{SE}$            | Single-ended receiver threshold                |                        | 0.8V            |             | 2.0V            |
| $V_{SEH}$           | Single-ended receiver hysteresis               |                        |                 | 200mV       |                 |
| $V_{OL}$            | Pad output low voltage                         |                        | 0V              |             | 0.3V            |
| $V_{OH}$            | Pad output high voltage                        |                        | 2.8V            |             | 3.6V            |
| $V_{CRS}$           | Differential output signal cross-point voltage |                        | 1.3V            |             | 2.0V            |
| $R_{PU}$            | External pull-up resistor                      | 1.5K $\Omega$ ±5%      | 1.425K $\Omega$ |             | 1.575K $\Omega$ |
| $R_{PD}$            | External pull-down resistor                    | 15K $\Omega$ ±5%       | 14.25K $\Omega$ |             | 15.75K $\Omega$ |
| $V_{TERM}$          | Termination voltage connected to $R_{PU}$      |                        | 3.0V            |             | 3.6V            |
| $Z_{DRV}^{\dagger}$ | Driver output resistance <sup>†</sup>          | Steady state drive     |                 | 10 $\Omega$ |                 |
| $C_{IN}$            | Transceiver pad capacitance                    | Pad to ground          |                 |             | 20pF            |

<sup>†</sup> Driver output resistance does not include the series resistance.

Table 8. High-Speed DC Electrical Specifications

| Symbol       | Parameter                                  | Conditions          | Min          | Nom | Max          |
|--------------|--------------------------------------------|---------------------|--------------|-----|--------------|
| $V_{HSDI}$   | High-speed differential input signal level | HSDP-HSDM           | 150mV        |     |              |
| $V_{HSSQ}$   | High-speed SQ detection threshold          | HSDP-HSDM           | 100mV        |     | 150mV        |
| $V_{HSCM}$   | High-speed common mode voltage range       |                     | -50mV        |     | 500mV        |
| $V_{HSOH}$   | High-speed data signaling high             |                     | 360mV        |     | 440mV        |
| $V_{HSOL}$   | High-speed data signaling high             |                     | -10mV        |     | 10mV         |
| $V_{CHIRPJ}$ | Chirp J level                              |                     | 700mV        |     | 1100mV       |
| $V_{CHIRPK}$ | Chirp K level                              |                     | -900mV       |     | -500mV       |
| $V_{HSTERM}$ | Termination voltage connected to $R_{PU}$  |                     | -10mV        |     | 10mV         |
| $Z_{HSDRV}$  | High-speed driver output resistance        | $45\Omega \pm 10\%$ | $40.5\Omega$ |     | $49.5\Omega$ |

## 4.2 AC Characteristics

### 4.2.1 Characterization Condition

Timing information of library *tpd973g\_usb2* is characterized under 3 operating conditions: worst-case, typical-case, and best-case conditions. The detail of each condition is listed in the following table:

Table 9. Characterization Condition

| Parameters         |             | Condition  |                 |           |
|--------------------|-------------|------------|-----------------|-----------|
|                    |             | Worst Case | Typical Case    | Best Case |
| <b>Process</b>     |             | Slow-Slow  | Typical-Typical | Fast-Fast |
| <b>Temperature</b> |             | 125°C      | 25°C            | 0°C       |
| <b>Voltage</b>     | <b>VDDA</b> | 3.0V       | 3.3V            | 3.6V      |
|                    | <b>VDDD</b> | 3.0V       | 3.3V            | 3.6V      |
|                    | <b>VDDL</b> | 1.62V      | 1.8V            | 1.98V     |

### 4.2.2 AC Electrical Specifications

Fig 10 shows the definition of USB signal rise time and fall time.

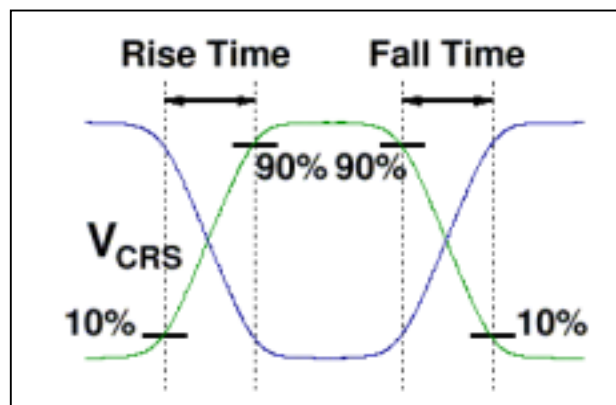


Fig. 10 Definition of rise time, fall time, and cross-point voltage ( $V_{CRS}$ )

Table 10 and Table 11 summarize the AC performance requirement of USB full-speed and high-speed transceivers according to the Chapter 7 of “Universal Serial Bus Specification Revision 2.0”. The USB transceiver of *tpd973g\_usb2* fully complies with the following specifications.



Table 10. Full-Speed AC Electrical Specifications

| Symbol        | Parameter                                 | Conditions                   | Min | Nom   | Max     |
|---------------|-------------------------------------------|------------------------------|-----|-------|---------|
| $T_{FR}$      | Full-speed driver rise time               | $C_L^{\S}=50\text{pF}$       | 4ns |       | 20ns    |
| $T_{FF}$      | Full-speed driver fall time               | $C_L=50\text{pF}$            | 4ns |       | 20ns    |
| $T_{FRFM}$    | Full-speed rise/fall time matching        | $T_{FRFM} = T_{FR} / T_{FF}$ | 90% |       | 111.11% |
| $I_{TAVD33F}$ | Full-speed TAVD33 current in steady state | Stand By                     |     | 0.5uA |         |
|               |                                           | Input Mode                   |     | 450uA |         |
|               |                                           | Output Mode                  |     | 450uA |         |

Table 11. High-Speed AC Electrical Specifications

| Symbol    | Parameter                                | Conditions           | Min                         | Nom | Max |
|-----------|------------------------------------------|----------------------|-----------------------------|-----|-----|
| $T_{HSR}$ | High-speed driver rise time              | $Z_{HSDRV}=45\Omega$ | 500ps                       |     |     |
| $T_{HSF}$ | High-speed driver fall time              | $Z_{HSDRV}=45\Omega$ | 500ps                       |     |     |
|           | High-speed driver waveform requirement   |                      | Eye diagram of template 1** |     |     |
|           | High-speed receiver waveform requirement |                      | Eye diagram of template 4†† |     |     |
|           | High-speed jitter requirement            | Data source end      | Eye diagram of template 1   |     |     |
|           |                                          | Receiver end         | Eye diagram of template 4   |     |     |

<sup>§</sup>  $C_L$  represents the equivalent capacitance applied in the AC test fixture of full/low-speed modes. For details of the test fixture, please refer to “*tpd973g\_usb2* silicon report v250b”.

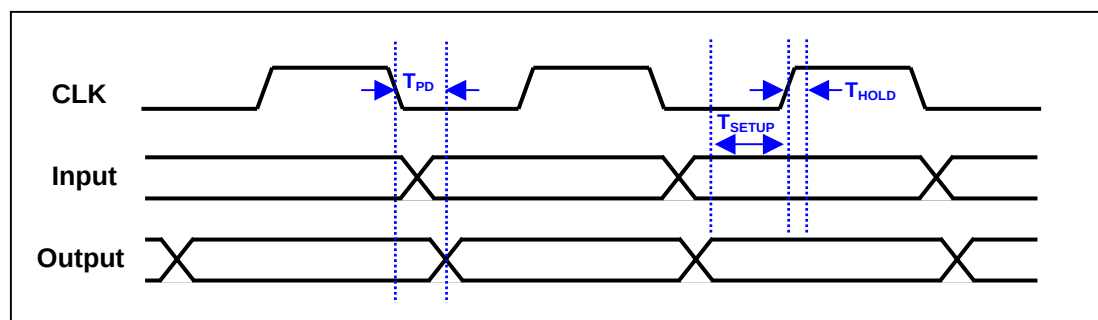
\*\* Check “Universal Serial Bus Specification Revision 2.0” page 133.

†† Check “Universal Serial Bus Specification Revision 2.0” page 136.

Table 12 shows the input setup/hold time requirement and output propagation delay of USB 8-bit and 16-bit digital interface signals. The definition of timing is shown in Fig 11. Note that for 8-bit case, CLK is 60MHz and for 16-bit case, CLK is 30MHz.

**Table 12. Setup/Hold Time and Propagation Delay of 8-bit/16-bit Interface**

| Symbol      | Parameter                                                 | Conditions                                        | Min | Max |
|-------------|-----------------------------------------------------------|---------------------------------------------------|-----|-----|
| $T_{SETUP}$ | Setup time of digital input signals <sup>‡‡</sup>         | Signal to CLK $\uparrow$                          | 4ns | -   |
| $T_{HOLD}$  | Hold time of digital input signals                        | Signal to CLK $\uparrow$                          | 0ns | -   |
| $T_{PD}$    | Propagation delay of digital output signals <sup>§§</sup> | CLK $\downarrow$ to signal<br>( $C_{LOAD}=10pF$ ) | -   | 3ns |



**Fig. 11 Definition of Input Setup/Hold Time and Output Propagation Delay**

<sup>‡‡</sup> Input signals include DATAIN [15:0], TXVAID, TXVALIDH, and BISTON.

<sup>§§</sup> Output signals include TXREADY, DATAOUT [15:0], RXVALID, RXVALIDH, RXACTIVE, RXERROR, LINESTATE [1:0], and BISTFAIL.



## 5. MACRO INFORMATION

In the following table, we show the macro information for users' reference.

Table 13. Macro Information

| Macro Name      | Bonding Pad No. | Macro Width (um) | Macro Height*** (um) | Power (uW/MHz) |     |     |
|-----------------|-----------------|------------------|----------------------|----------------|-----|-----|
|                 |                 |                  |                      | WC             | TC  | BC  |
| <b>PUSB20TM</b> | 11              | 1277.81          | 731.52               | TBD            | TBD | TBD |

\*\*\* The macro height includes the I/O cells but NOT the bonding pad. For linear bonding, the macro height with bonding pad is 821.32um.



## 6. DESIGN KIT SUPPORT

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Various kinds of design kits are delivered along with the *tpd973g\_usb2* library. These design kits are modeled for most popular EDA tools to facilitate the design flow in integrating with other libraries or IPs. Table 14 lists all the design kits supported and their contents.

Table 14. List of the Delivered Design Kits

| Design Kit       | Contents                                                |
|------------------|---------------------------------------------------------|
| <i>rln</i>       | Release note                                            |
| <i>syn</i>       | Synopsys <sup>TM</sup> db, lib files                    |
| <i>vlg</i>       | Verilog <sup>TM</sup> simulation model                  |
| <i>apf / apt</i> | Apollo <sup>TM</sup> layout frame view and cell view    |
| <i>sef</i>       | Silicon Ensemble <sup>TM</sup> lef and technology files |
| <i>gds</i>       | GDSII layout database                                   |
| <i>spi</i>       | LVS netlist in CDL <sup>TM</sup> format                 |
| <i>doc</i>       | Datasheet and application notes                         |



## 7. CONTACT US

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TSMC specialty I/O library *tpd973g\_usb2* is released after passing the standard quality assurance (QA) procedure. The specific checking environment, including the EDA tool version and technology files are included in the release database. If users found any errors or encountered any problem with these models, please check the “release note” and “designkit.info” for the known issues first. Users should also make sure that their tool version, design rule manual, and DRC/LVS command files all comply with TSMC’s QA condition. For advanced issues, please contact TSMC’s local FAE.